Information Management for CS Introduction

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Slides based on Prof.dr.ir. Hajo Reijers

Relevance Information Systems

- For all organizations of some size, information systems are an essential element in their operations
- To many organizations their information systems are just as essential as their transformation system
- To some organizations their transformation systems are dominantly IT-driven

Course organization (1/4)

Responsible lecturers:

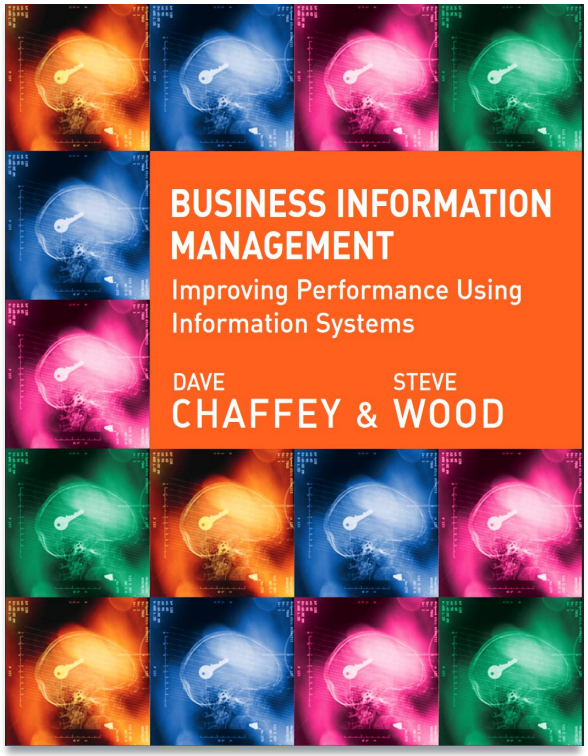
Ilias Gerostathopoulos

Abhishek Iyer

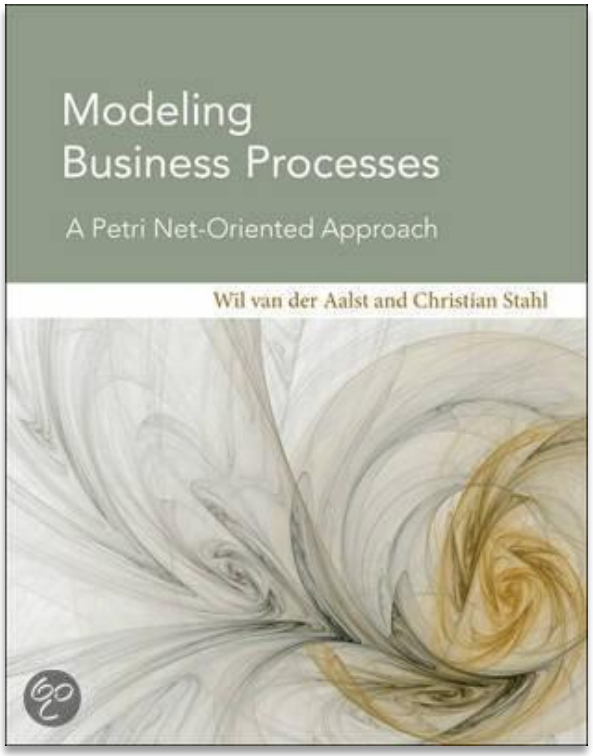
Teaching assistants:

Sara Abesová
George Dedempilis
Panagiotis Fotopoulos
Lucia Sikulova
Ioana Song
Karolina Hajkova

Course organization (2/4)

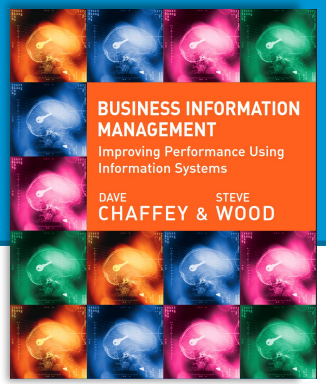


- First half
- Lectures
- Intermediate exam: April 21



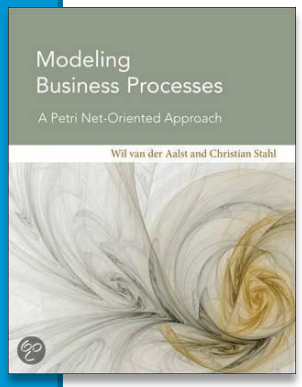
- Second half
- Lectures + Seminars + Pass/Fail Assignments
- Group project: June 1
- Final exam: May 29

Course organization (3/4)



Information Systems

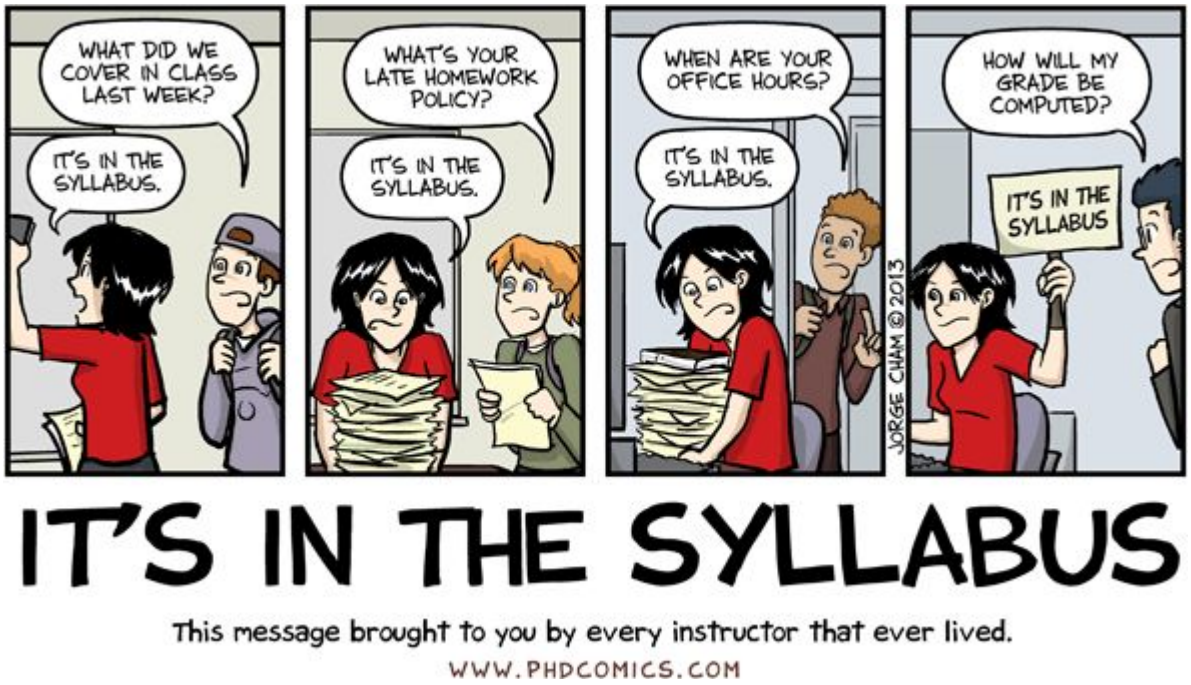
Process modeling



Course organization (4/4)

- The Course Guide (Syllabus) on Canvas contains detailed information on the course:
  - Course structure
  - Study schedule
  - Grading
  - Etc.

- Read it: we think it's worth the effort.



Course schedule

TYPE	DATE AND TIME	TOPIC
Lecture 1	**April 2nd, 11:00-12:45**	Course outline & Information System strategy
Lecture 2	**April 9th, 11:00-12:45**	Information Systems and the Internet
Lecture 3	**April 14th, 11:00-12:45**	Development of Information Systems
Lecture 4	**April 16th, 11:00-12:45**	QnA Session
Intermediate exam	**April 21st, 12:15-13:45**	Intermediate exam on first part of the course (first three weeks)
Lecture 5	**April 23rd, 11::00-12:15**	Transition systems + introduction to Petri Nets
Seminar 1	**May 6th, 17:30-19:15**	Transition systems
Lecture 6	**May 7th, 11:00-12:45**	Petri Nets
Resit for Intermediate exam	**May 12th, 12:15-13:45**	Resit for Intermediate exam on first part of the course (first three weeks)
Seminar 2	**May 13th, 17:30-19:15**	Petri Nets
Lecture 7	**May 19th, 11:00-12:45**	Workflow nets & analysis
Seminar 3	**May 20th, 11:00-12:45**	Workflow nets & analysis
Lecture 8	**May 21st, 11:00-12:45**	Real-life Petri Nets examples, course recap
Final Exam	**May 29th, 18:45-21:30**	Exam covering the second part of the course (weeks 4-7)
Resit for Final Exam	**July 2nd, 12:15-15:00**	Resit for Final exam covering the second part of the course (weeks 4-7)

What do we expect from you?

- Active participation in lectures + seminars
- Timely submission of Pass/Fail Assignments
- Keep up with the material: *weekly study instructions*
- Interact through Canvas:
  - Take note of messages/notifications/updates
  - Put your questions there
  - Put your answers there as well!
- No mandatory presence, yet practice is indispensable

Study load

Week	Total number of hours	Lecture hours	Seminar hours	Exam hours	Homework hours	Project work hours	Self-study hours
1	18	2					16
2	18	2					16
3	18	4					14
4	18	2		2			14
5	20	2	2		6		10
6	22	2	2		6	4	8
7	22	2	2		6	4	8
8	26			2		24	
Total	*162*	*16*	*6*		*24*	*48*	*72*

Lecture 01 - Study instructions

BUSINESS INFORMATION MANAGEMENT - Improving Performance using Information Systems by DAVE CHAFFEY & STEVE WOOD

Essential (exam material):

- Chapter 1 - Introduction to Information Management
  - (Information in today's organization)
  - (Fundamentals of Business Information Management)
  - (Information Resources)
- Chapter 4 - Information management strategy
  - (Developing an information strategy management) (only go through the Strategy Process sub-section)
- Chapter 6 - Information System Strategy
  - (What is IS Strategy) (IT Governance sub-section is not for exam)
  - (Defining a process for IS strategy development) (Responsibilities and controls for IS strategy sub-section is not for exam)
  - (The stages of IS strategy development)

Background material (read):

- Chapter 1 – Introduction to Information Management
  - (Introduction)
  - (Information in today's world)
- Chapter 4 - Information management strategy
  - (Why is an information management strategy needed?)
  - (The relationship between strategies for managing information-related resources)
- Chapter 6 - Information system strategy
  - (Strategic Situation Analysis)

Study advice:

- **You do not need to go through the case studies or the research insights provided in the chapters, they will not be part of the exam questions.**
- For Chapter 1, you should particularly understand how the role of information systems has changed over the past decades and of which importance information systems are today.
- For Chapter 4, you should be able to explain why an information management strategy is necessary and how the strategy process works.
- For Chapter 6, you should be able to understand the different types and stages of IS strategy.
- It is highly recommended to read the background material. It complements the concepts discussed in the lecture.

Grading

1. Intermediate exam: 40%
2. Group project: 40%
3. Final Exam: 20%

To obtain your grade you need to also pass all 3 pass/fail individual Canvas assignments

Plan for lectures

1. Information and Strategic planning
2. Information Systems and the Internet
3. Information Systems Development
4. QnA Session
5. Business Processes and Information Systems
6. Petri Nets
7. Analysis with Petri Nets
8. Real life/complex Petri Nets

Agenda for this lecture

1. Information and Information System(IS)
2. Strategic Planning
3. Developing an IS Strategy

Information and IS

Today's Business Challenges

- Intense competition and environmental changes
- Increase of amount of knowledge that affect your business -> Knowledge Management
- Fast change of business environment: Time-based competition

Just Testing: Google Users May See Up To A Dozen Experiments

Google's Chief Economist Hal Varian said during a Google Tech Talk on October 22, 2012 that Google users may see a dozen or more Google experiments per Google visit. About 26 minutes into the Google Tech Talk, Hal said, "any time you access Google, you probably are in a dozen or more experiments." The talk [...]

Barry Schwartz on December 5, 2012 at 10:22 am

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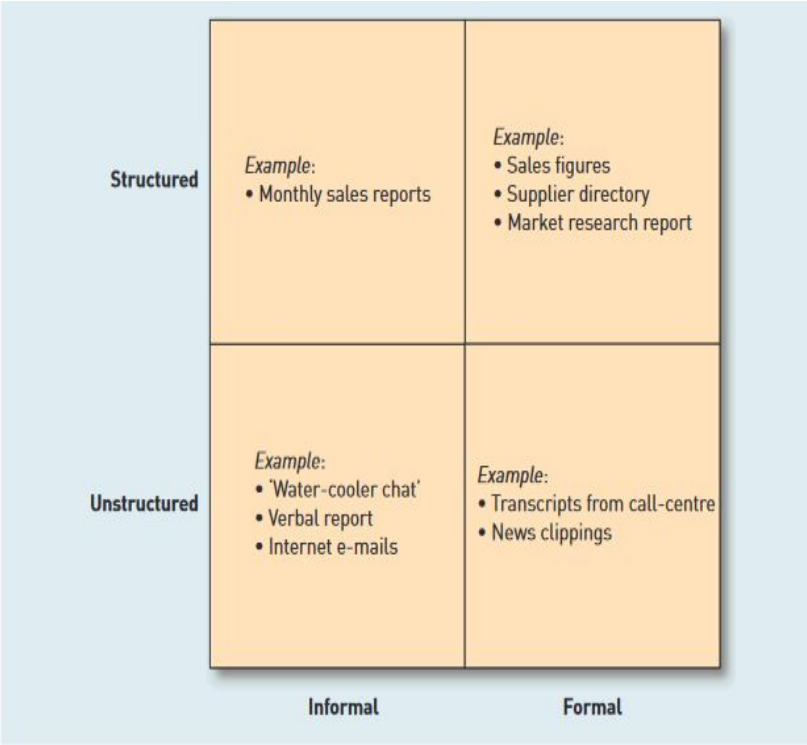
Information in today's organization

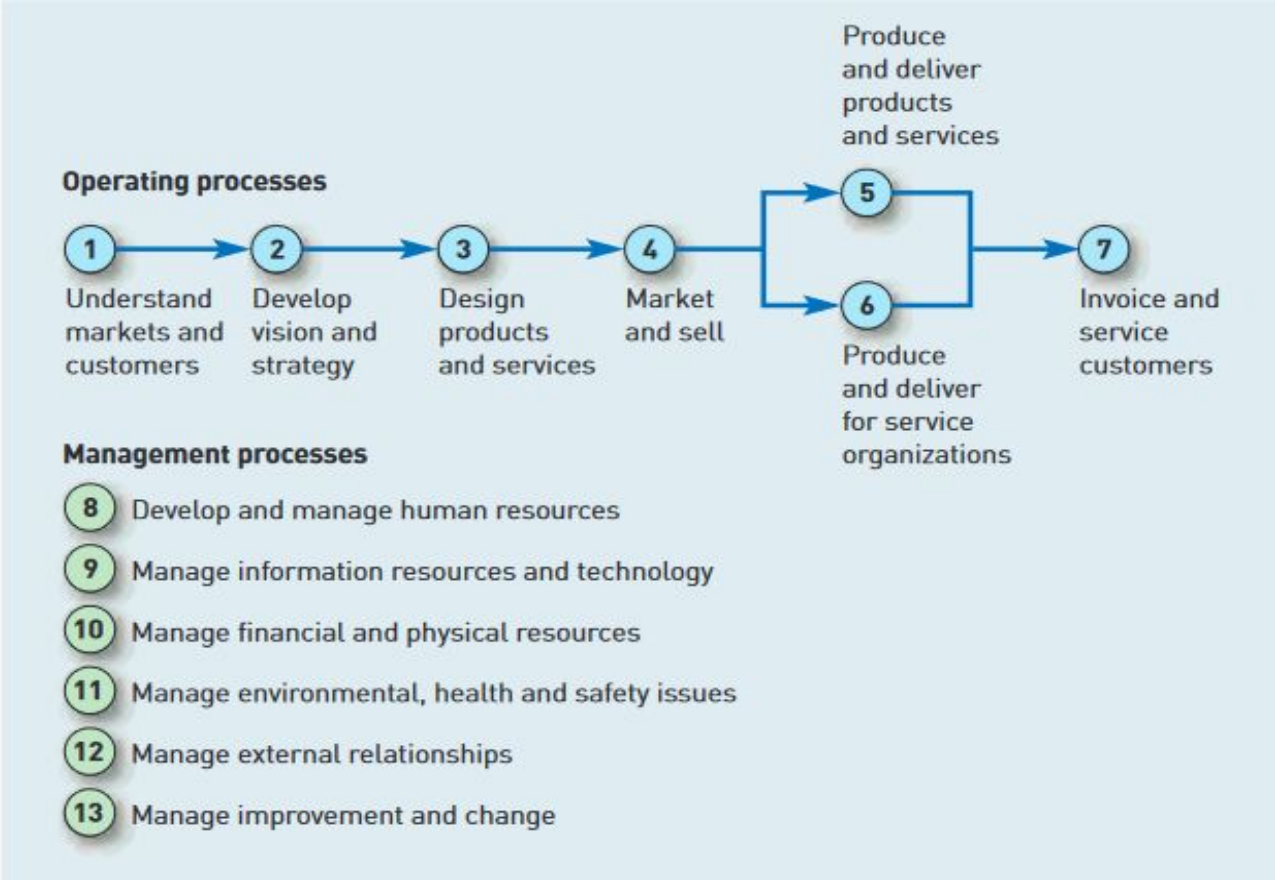
- Increasing dependence on information and the systems that deliver this information
- Increasing vulnerabilities and a wide spectrum of threats
- Scale and cost of the current and future investments in information and information systems
- Potential for technologies to dramatically change organizations and business practices, create new opportunities and reduce costs.

Information Types and Sources

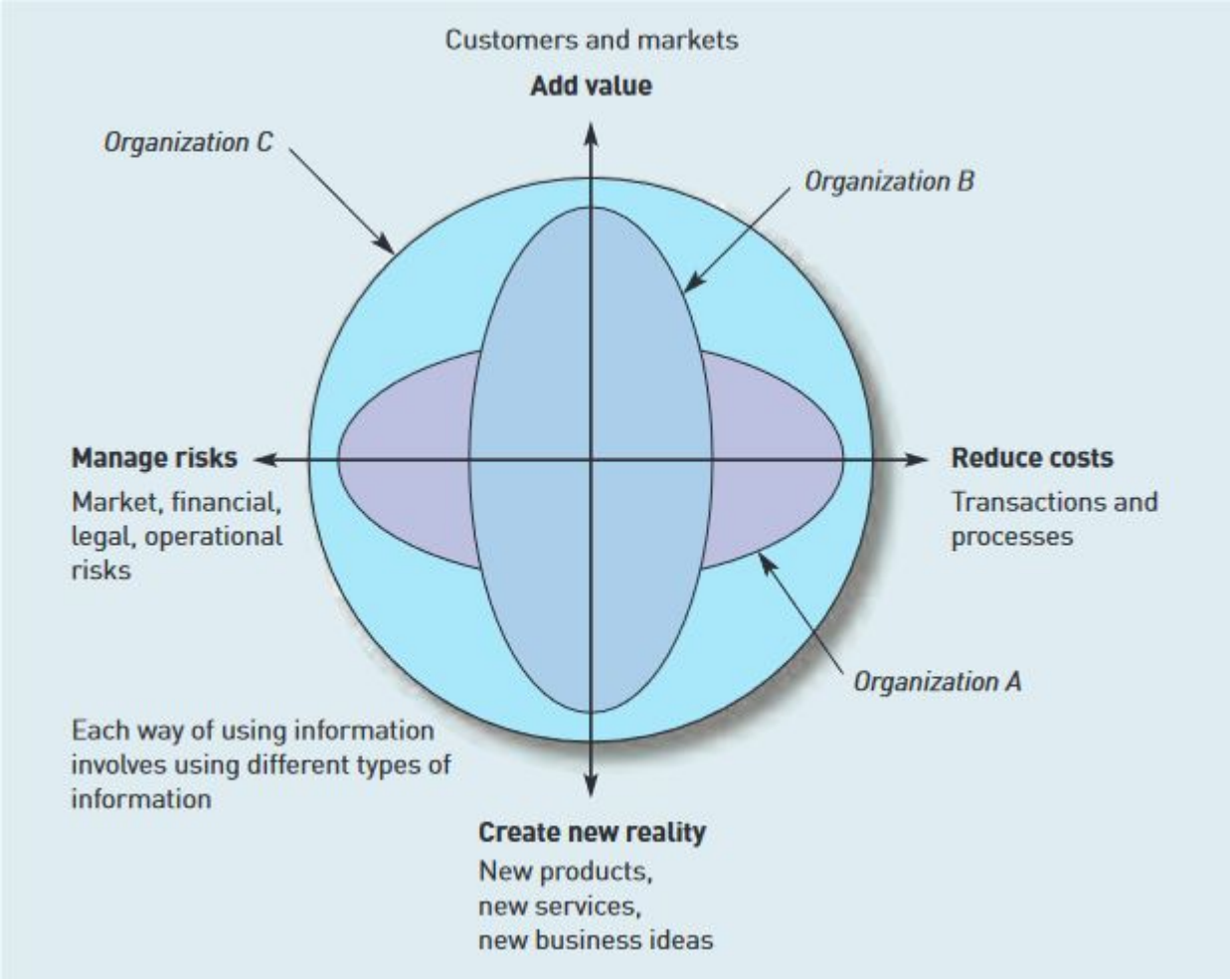
- Structured information – presented in reports, tables and graphs
- Unstructured information – delivered verbally or on an ad-hoc basis
- Formal information – part of established reporting and communication
- Informal information – ad-hoc communication such as conversations or e-mail.



Using information to support processes



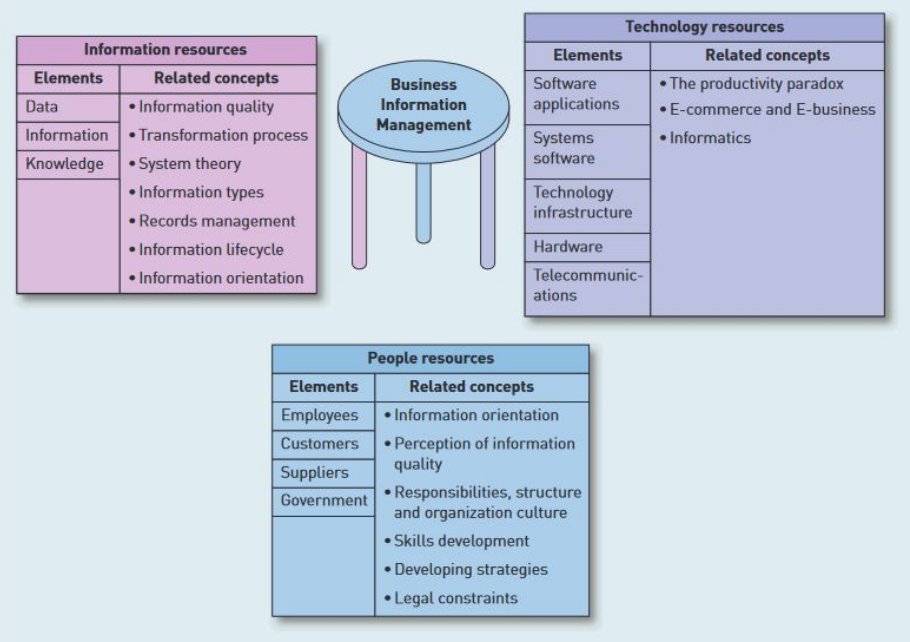
Using information to create value



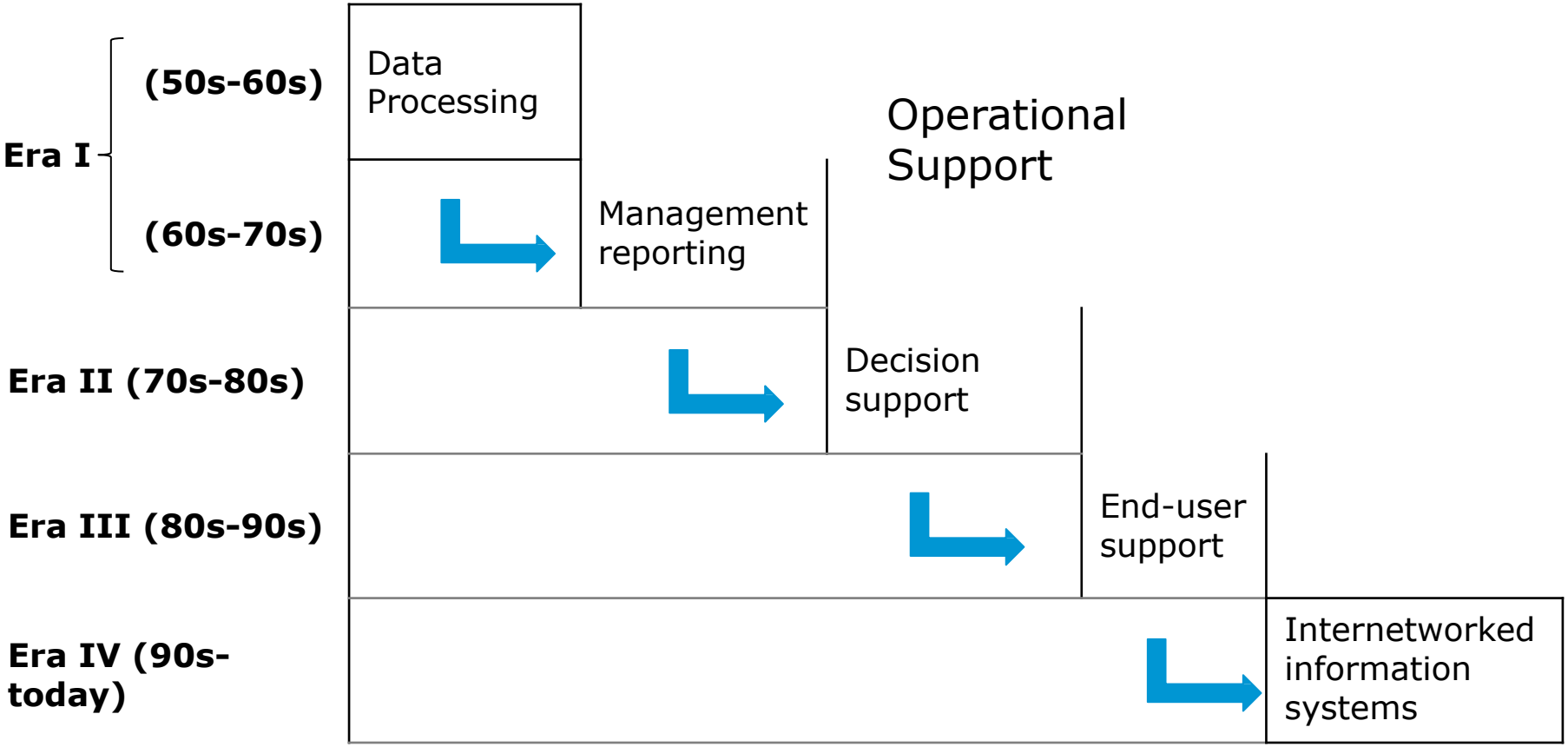
Fundamentals of Business Information Management

- The three strands of business information management are:
  - Information Resources
  - People Resources
  - Technology Resources

The process of managing information as a strategic resource for improving organizational performance. This process involves developing strategies and introducing systems and controls to improve information quality to deliver value.



Historical Development of Information Systems



Use of Information Systems in Businesses

Executive Information Systems

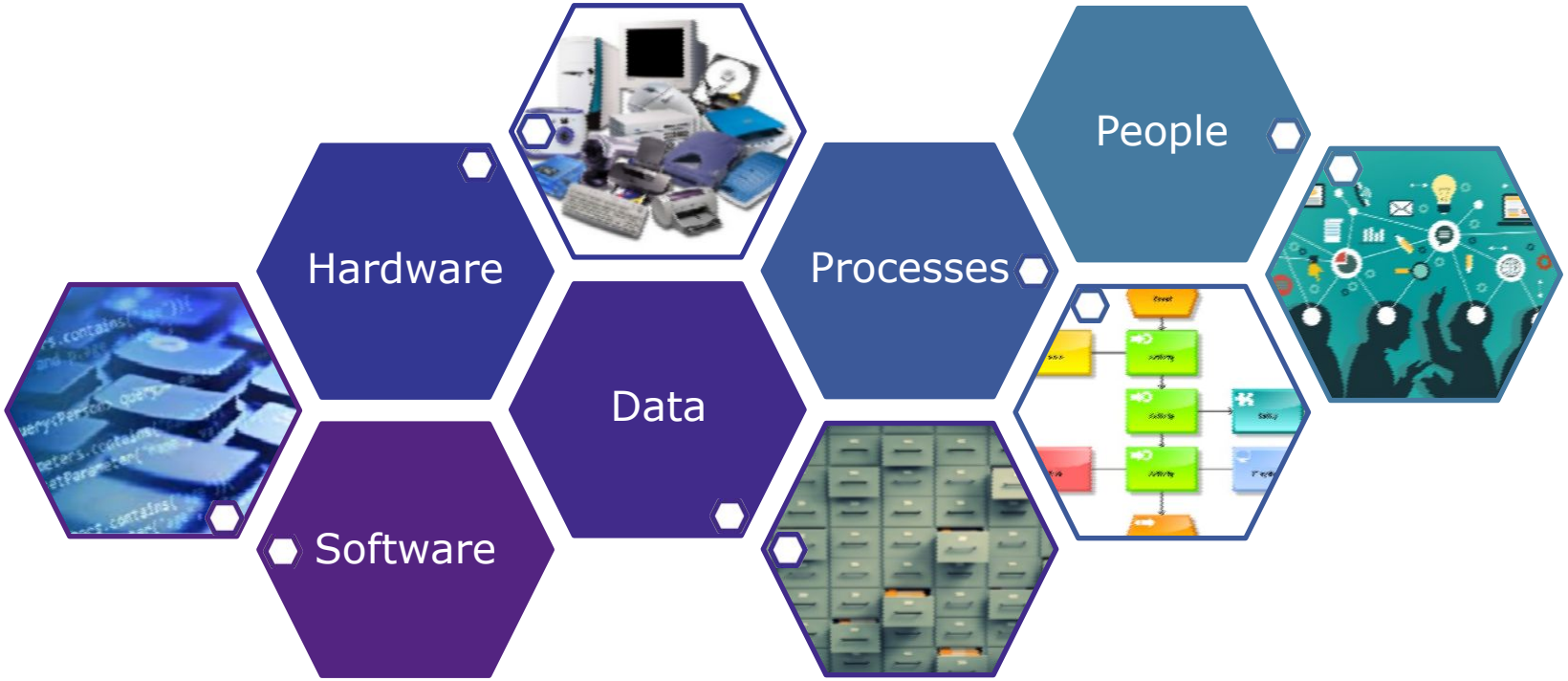
Management Information Systems
Decision support systems

Transaction Processing Systems
Process control systems

Information Characteristics at Each Level

Management level	Information					
	Time period	Frequency	Source	Certainty	Scope	Detail
Strategic	Wide	Infrequent	External	Less certain	Wide	Summarised
Tactical	↔	↔	↔	↔	↔	↔
Operational	Narrow	Frequent	Internal	More certain	Narrow	Detailed

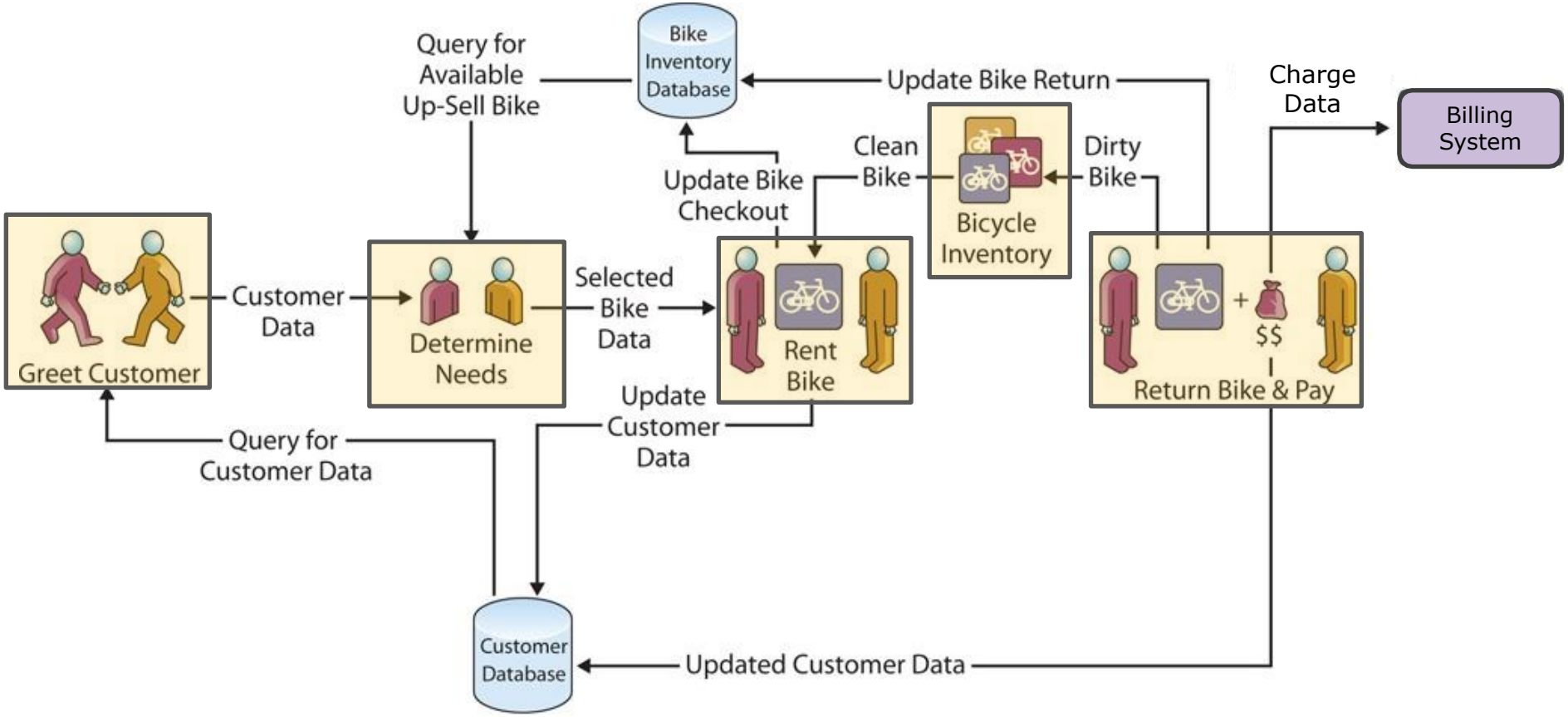
Components of an IS



Scope of Processes

Scope	Characteristics	Mix of actors	Frequency of occurrence	Examples	Supporting IS
Operational	Common, routine, day-to-day	More computers than other processes	High	Procurement Assemble product Track orders Pay bills Order	TPS
Tactical	Allocation and use of resources	Mix	Medium	Manage inventory Schedule maintenance Evaluate complaint patterns Assess seasonal promotions	MIS
Strategic	Broad-scope, organizational issues	More people than other processes	Low	Evaluate potential suppliers Launch new product Decide on new location Corporate budgeting	EIS

What are the components of this Bike Rental IS?



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Hardware Software Data Processes People

What are the components of this Bike Rental IS?

Hardware	Bikes (clean and dirty), servers, personal computers, hand terminals
Software	Bike rental management (customer query and update, bike search, bike return), billing
Data	Customer data, bike data, bike rental data, charge data
Processes	Bike rental (bike search and select, bike return), billing, bike maintenance, retrieve & update customer data
People	Customer, shop employees (rental and maintenance)

Classify the following decisions (strategic-tactical-operational)

a) Should we add more workers to a task to cope with an urgent order?
b) Should we launch an advertising campaign?
c) What new markets should we move into?
d) At what level should we set the budget for next year?
e) Does this customer qualify for a discount on a large order?
f) How should we deal with a takeover bid?
g) Should we expand abroad?
h) Should we take a short-term loan to help our current cash flow position?
i) What to do with a faulty machine?

Strategic Planning and IS

The need for strategic planning

- Every organization has an overall direction
- Day-to-day activities are performed to achieve the main purpose of the organization
- There is a need for continuous strategic planning due to
  - Ever-changing internal and external factors necessitating resource allocations
  - Need for subsystem coordination
  - Expectations of different stakeholders (owners, customers, employees, …)

The IS Strategy Process

There are five main stages in developing and implementing strategy. These five stages answer these questions.

1) ‘Where are we now?’ – the situation analysis.
2) ‘Where do we want to be?’ – the vision and objectives.
3) ‘How are we going to get there?’ – the strategy.
4) ‘How do we introduce the changes?’ – the implementation of the strategy.
5) ‘How are we doing?’ – the monitoring and control of strategy’.

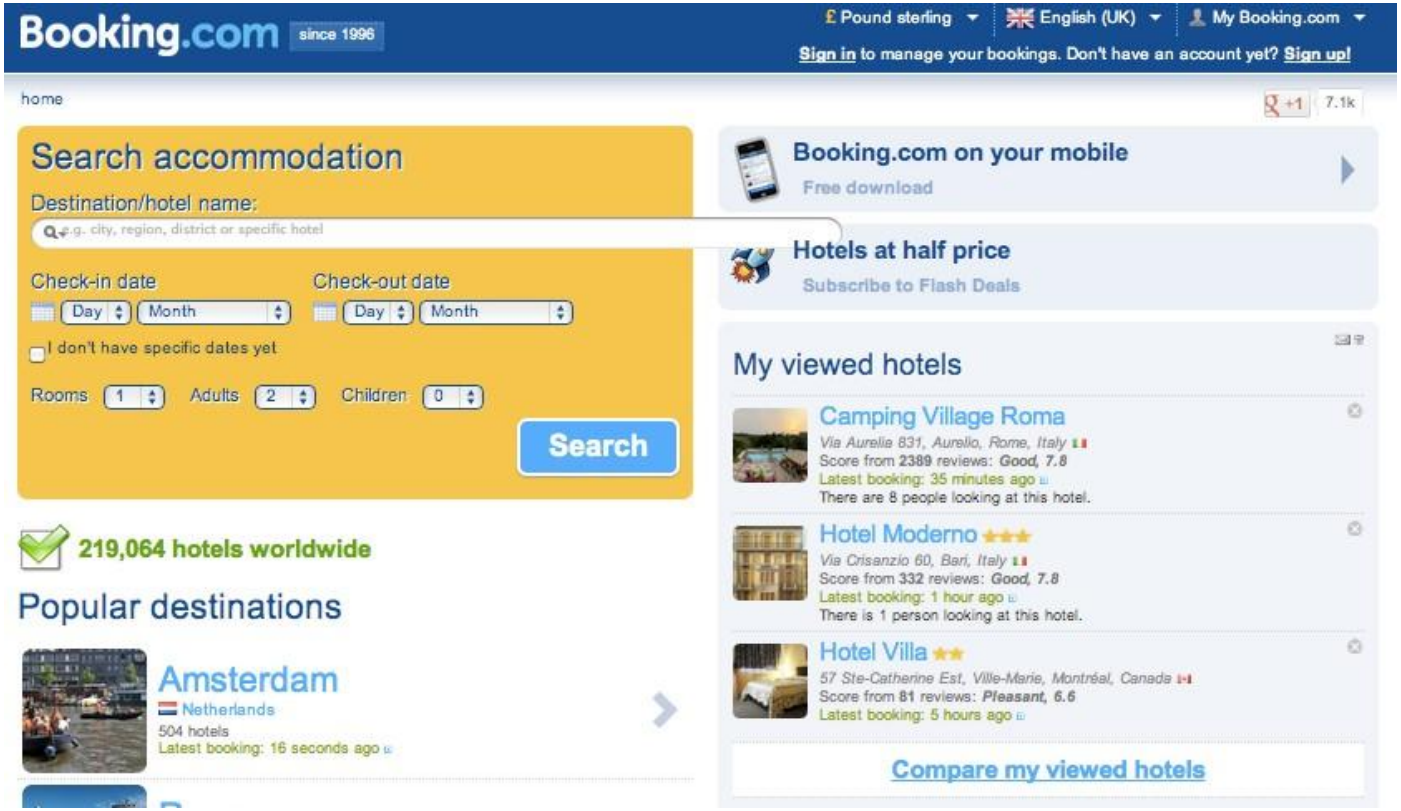
Strategic Relevance of IS (1/3)

- A strategic system alters the way an organization does business
- Some systems offer a company a clear competitive advantage: higher profits or increased market share
- Most systems enable a company to be an *effective competitor*

Strategic Relevance of IS (2/3)

- Rapid diffusion of technological change makes it difficult to maintain a competitive advantage
- Hence, strategic development of IS is
  - a dynamic capability of an organization
  - not a static attribute

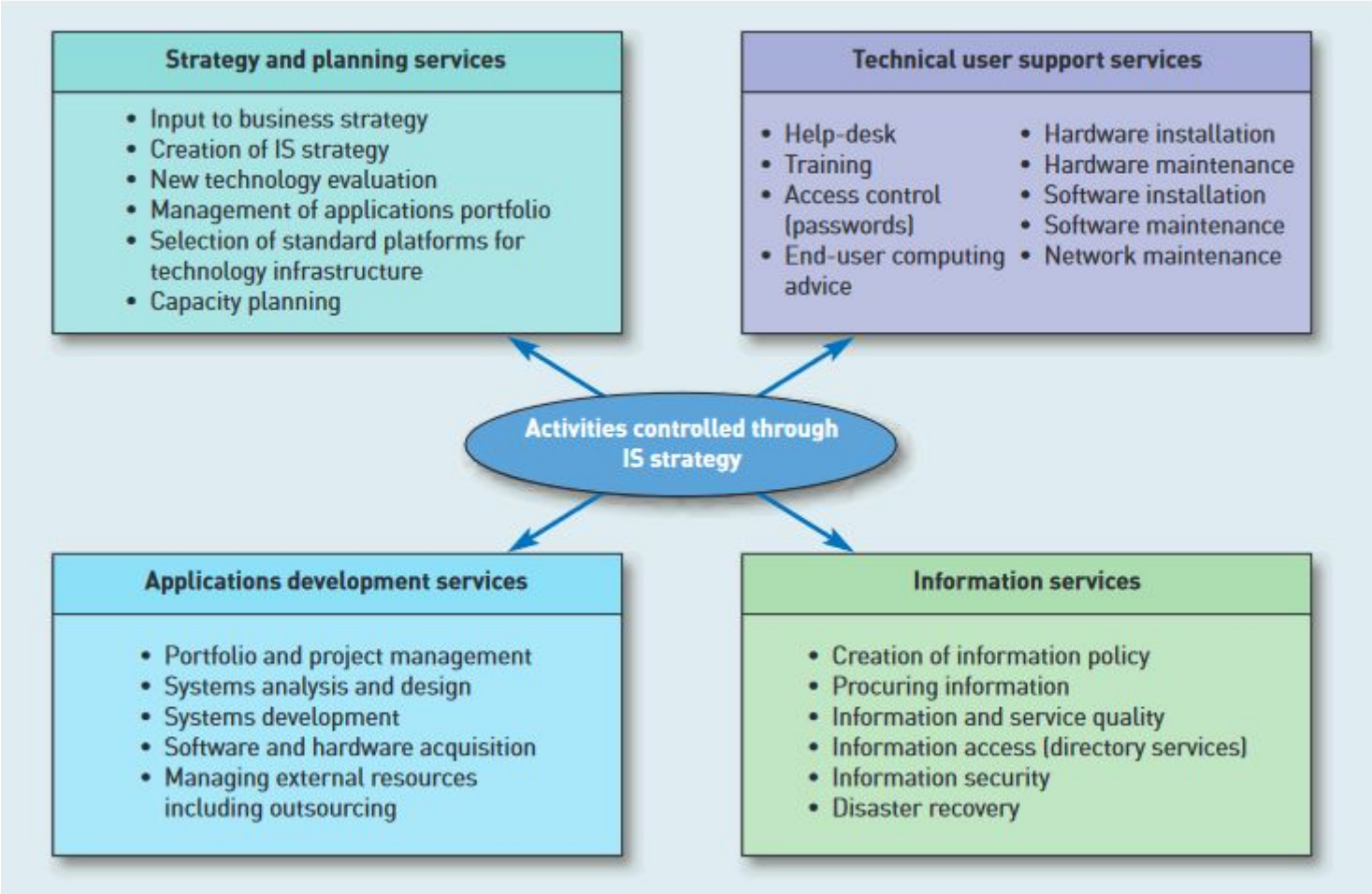
Strategic Relevance of IS (3/3)



IS strategy activities (1/2)

- IS strategy have been usefully grouped into four different types of services by Ward and Peppard (2002):
  - Strategy and planning services.
  - Applications development services.
  - Application and technical management services.
  - Technology, delivery and maintenance services.

IS strategy activities (2/2)



Developing an IS Strategy

Defining a process for IS strategy development

- The IS strategy development process should have these characteristics:
  - *Achieves alignment of IS strategy with business strategy while identifying competitive opportunities available through IS*
  - *Simplicity through well-defined stages*
  - *Continuous process with evaluation and improvement built-in*
  - *Flexibility*

Long-term vs short-term scope of IS strategy

Long-term IS Strategy (3-5 years):

- Specifies broad strategic approaches over an extended period.
- Considers future technologies and market trends.
- Allows for staged implementation of technologies, such as ecommerce.
- Enables investment in information systems to be spread over time.
- Considers potential changes in market demand and technology costs.

Short-term IS Strategy (6-12 months):

- Focuses on detailed strategies for immediate implementation.
- Addresses specific challenges and opportunities within a shorter timeframe.
- Adapts to rapidly changing market conditions and technological advancements.
- Provides agility in responding to emerging trends and customer demands.
- Aligns with long-term goals while accommodating short-term priorities.

Top-down vs Bottom-up IS strategy

Top-down IS Strategy:

- Begins with identifying overarching business objectives, such as increasing market share or improving customer satisfaction.
- Evaluates which information systems can best support these high-level goals.
- Focuses on aligning IS initiatives with the organization's strategic direction

Bottom-up IS Strategy:

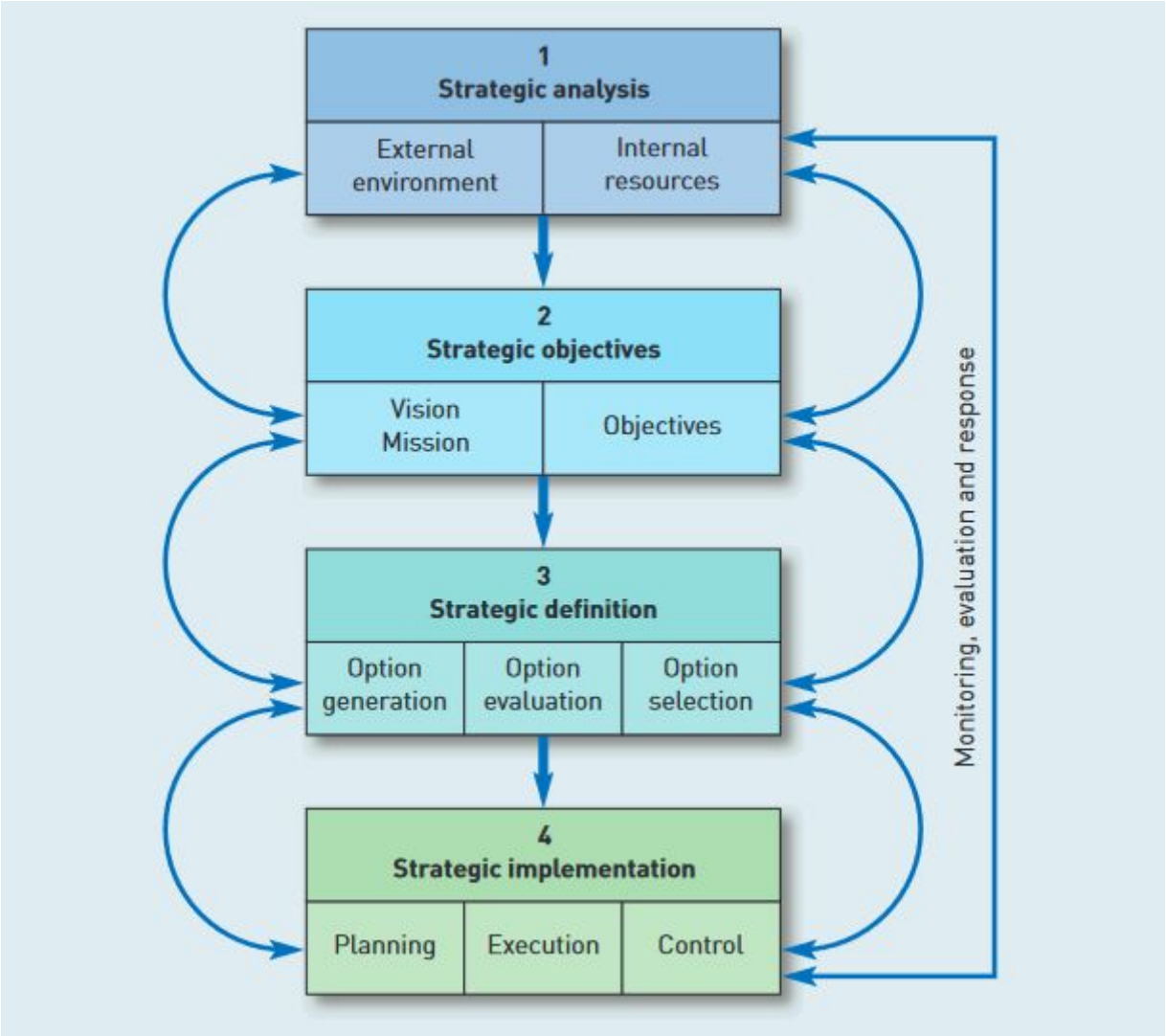
- Starts with individual departmental or functional needs within the organization.
- Emphasizes the impact of IS based on specific requests or requirements from different departments.
- May result in a diverse array of systems tailored to different operational areas.

Five approaches to IS strategy development

The five main approaches to the IS strategy development process are :

- Business-led approach
- Method-driven approach
- Administrative approach
- Technological approach
- Organizational approach

The stages of IS strategy development



Summary

- Information quality and the use of information systems is very important in current business settings
- Information systems may create competitive advantage
- An existing advantage, however, may quickly disappear
- Strategy plays a crucial role in getting the most of the Information System
- Different requirements call for different strategy

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